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1  LIBRARY ieee;
2  USE ieee.std_logic_1164.ALL;
3  USE ieee.numeric_std.ALL;
4
5  ENTITY s22_b IS
6      PORT (
7          clk : IN STD_LOGIC;
8          y : OUT STD_LOGIC_VECTOR(3 DOWNTO 0)
9      );
10 END ENTITY;
11
12 ARCHITECTURE behavioral OF s22_b IS
13     SIGNAL q : STD_LOGIC_VECTOR(3 DOWNTO 0) := "1101";
14     SIGNAL yin : STD_LOGIC_VECTOR(3 DOWNTO 0) := (OTHERS => '0');
15 BEGIN
16
17     -- sequential process
18     seq_p : PROCESS (clk)
19     BEGIN
20         IF rising_edge(clk) THEN
21             -- the data shift inside the register
22             q <= q(0) & q(3 DOWNTO 1);
23             IF q = "1110" THEN
24                 yin <= (OTHERS => '0');
25             ELSE
26                 yin <= q;
27             END IF;
28         END IF;
29     END PROCESS;
30
31     y <= yin;
32
33 END ARCHITECTURE behavioral;

```